

# TANCET MCA

## Original Question Paper - 2001

Directions (Qns. 1 to 20): Choose the best answer to the given questions.

1. A works twice as fast as B. If B can complete a piece of work independently in 12 days, find in how many days A and B together can complete this work?  
(A) 18 days  
(B) 4 days  
(C) 6 days  
(D) 8 days
2. One manufacturing plant built 150 tanks in the last six months in 1975. This was an increase of 150% over the number built in the preceding six months. Find the number of tanks built in the preceding six months.  
(A) 100  
(B) 50  
(C) 0  
(D) 60
3. 961 is divisible by  
(A) 21  
(B) 31  
(C) 11  
(D) 41
4. If a sum is divided between A and B such that 3 times A's share is the same as 2 times B's share, what part of the sum A's share?  
(A)  $2/3$   
(B)  $3/5$   
(C)  $2/5$   
(D)  $1/3$
5. The value of X for which  $1 + \sqrt{25/144} = 1 + x/12$  is  
(A) 1  
(B) 5  
(C) 12  
(D) 13
6. By how much is four-fifth of 70 lesser than five-seventh of 1121?  
(A) 42  
(B) 2  
(C) 24  
(D) 40
7. A shopkeeper increased the price of an article first by 25% and then by 20%. What is the total percentage of increase?  
(A) 40  
(B) 45  
(C) 50  
(D) 55
8. The area of the ellipse  $x^2/a^2 + y^2/b^2 = 1$  is given by  
(A)  $(\pi/2) ab$   
(B)  $\pi ab$   
(C)  $ab$   
(D)  $(\pi/3)ab$
9. The average age of 30 boys in VI class is 16. If the teacher's age is included, the average increases by one. What is the teacher's age?  
(A) 37 years  
(B) 30 years  
(C) 47 years  
(D) 57 years
10. An album has 20 pages, each page measuring  $20 \times 15$  cm. A boy purchases daily six stamps measuring  $2 \times 2.5$  cm. In how many days the album will be completely pasted with stamps?  
(A) 160  
(B) 180  
(C) 196  
(D) 200
11. If  $7x - 5y = 13$  and  $2x - 7y = 26$  then  $5x + 2y =$   
(A) -39  
(B) -13  
(C) 13  
(D) 39

12. If  $6x^{2k-4} + 7x^{k-2} + x^{k-3} = (6x+1)(x+1)$  then the value of  $k$  must be  
 (A) 1  
 (B) 2  
 (C) 3  
 (D) 4
13. Raj and Ram are partners in a business. Raj invests Rs. 5,000 for 5 months and Ram invests Rs. 6,000 for 6 months. If the profit is Rs. 610, then Ram's share in the profit is  
 (A) Rs. 250  
 (B) Rs. 360  
 (C) Rs. 520  
 (D) Rs. 630
14. If  $9x+5=23$ , the numerical value of  $18x+5$  is  
 (A) 46  
 (B) 41  
 (C) 32  
 (D)  $23+9x$
15. A man buys a T.V. set which is listed for Rs. 18,000 at a 10% discount. He gets a further discount of 2% on the balance because he paid cash. The amount he actually paid for the T.V. set is  
 (A) Rs. 16,256  
 (B) Rs. 16,200  
 (C) Rs. 15,876  
 (D) Rs. 15,840
16. If  $r+q=s$  and  $r+q=w$ , what does  $w$  equal in terms of  $r$  and  $q$ ?  
 (A)  $2r+q$   
 (B)  $2q+r$   
 (C)  $2r+2q$   
 (D)  $r+q+s$
17. The difference of 2 numbers is 3. Twice the first is 25% more than the second. Find the larger of the two  
 (A) 10  
 (B) -8  
 (C) -5  
 (D) 8
18. The area of the rectangle is thrice that of a square. Length of the rectangle is 40cm and breadth of the rectangle is

$3/2$  times that of the side of the square. What is the side of the square in cm?

- (A) 60  
 (B) 20  
 (C) 30  
 (D) 15
19. A gives B 40% of the money A has with himself. Then B gets Rs. 200. How many rupees does B get from A?  
 (A) 80  
 (B) 200  
 (C) 300  
 (D) 160
20. A purse contains 2 silver and 4 copper coins and a second purse contains 4 silver and 4 copper coins. If a coin is selected at random from one of the two purses, what is the probability that it is a silver coin?  
 (A)  $4/3$   
 (B)  $5/12$   
 (C)  $1/3$   
 (D)  $1/5$

*Directions (Qns. 21 to 25): Answer the questions based on the following passage.*

Below is given a passage followed by several possible inferences which can be drawn from the facts stated in passage. You have to examine each inference separately in the context of the passage and decide upon its degree of truth or falsity.

- (A) Mark answer "DEFINITELY TRUE", if you think the inference properly follows from the statements of facts given.  
 (B) Mark answer "PROBABLY TRUE", if you think the inference may be true in the light of the facts given, but not definitely true.  
 (C) Mark answer "CANNOT BE DETERMINED BY THE GIVEN INFORMATION", if you think from the facts it cannot be said whether the inference is likely to be true or false.  
 (D) Mark answer "PROBABLY FALSE", if you think the inference is probably false in the

light of the facts given though not definitely false.

- (E) Mark answer "DEFINITELY FALSE", if you think the inference cannot possibly be drawn from the facts given or it contradicts the five facts.

### PASSAGE

"Steel production in India has always been burdened with costs much higher than in comparable plants overseas. On the one hand custom duty on almost all steel imports has been reduced on the other excise duty has been raised. Domestic steel products are now uncompetitive which in turn has led to the demand that both the custom and excise rates should be reviewed to prevent sickness in the industry"

For the next five questions, pick the correct answer which exactly follows from the information given above.

21. Foreign steel has become cheaper than domestic steel.  
(A) Probably true  
(B) Definitely true  
(C) Cannot be determined by the given information  
(D) Definitely false
22. The government is considering lowering of excise duty on steel products.  
(A) Cannot be determined by the given information  
(B) Definitely false  
(C) Probably false  
(D) Probably true
23. The cost of production of steel in other countries are less than India  
(A) Definitely false  
(B) Definitely true  
(C) Probably true  
(D) Probably false
24. The steel industry in general is making losses as a fall out of the government policy

- (A) Probably false  
(B) Definitely true  
(C) Probably true  
(D) Definitely false

25. India can compete with the foreign countries in regard to the price of steel with a favourable government policy.

- (A) Definitely true  
(B) Cannot be determined by the five information  
(C) Probably false  
(D) Probably true

Directions (Qns. 26 to 34): Choose the correct answer to the following questions.

26. If NQXG means LOVE, then ECMG means  
(A) Good  
(B) Calm  
(C) Cake  
(D) Calf
27. Pointing to a picture in a photograph, a woman said: "this man is the father of the brother of my mother's one of the two son-in-law". How is the man related to the woman?  
(A) Father  
(B) Father in-law  
(C) Brother  
(D) Husband
28. If 5 men can do a piece of work in 6 days and 10 women can do it in 5 days, in how many days can 3 men and 5 women do the same piece of work?  
(A) 3 days  
(B) 6 days  
(C) 5 days  
(D) 4 days
29. The average age of 28 students in one section was 5 years and 7 months. One student aged exactly 5 years and 7 months withdraw and another boy was taken from the waiting list. This change reduced the average age of students to 5 years and  $6\frac{3}{4}$  months. The age of the new boy therefore is



- (A) 4 years 11 months  
(B) 4 years 11 1/4 months  
(C) 5 years 0 month  
(D) 5 years 1 1/4 month
30. Instead of multiplying a number by  $\frac{2}{5}$  as required, a boy divided it by  $\frac{2}{5}$ . By this, his result was  
(A)  $\frac{2}{5}$  times the true answer  
(B)  $\frac{5}{2}$  times the true answer  
(C)  $\frac{2}{25}$  times the true answer  
(D)  $\frac{25}{4}$  times the true answer
31. The product of 3 number is 750 and their ratio is 1:2:3. What is the sum of the squares of three numbers?  
(A) 375  
(B) 350  
(C) 400  
(D) 450
32. The coordinates of the centre of the conic  $x^2 - 3xy + y^2 + 1 = 0$  is given by  
(A) (1, 2)  
(B) (0, 0)  
(C) (1, 1)  
(D) (0, 2)
33. Divide Rs. 1,800 between Ram, Shyam and Mohan so that for every rupee that Ram gets Shyam gets 2 rupees and Mohan gets 3 rupees  
(A) 400, 600, 800  
(B) 300, 600, 900  
(C) 500, 400, 900  
(D) 600, 500, 700
34. Consider the following two statements  
All classes are glasses  
All brasses are glasses  
Consider the following conclusions  
I Some classes are brasses  
II Some brasses are glasses  
III Some brasses are classes  
IV Some classes are glasses  
Which of the given conclusions follow in the right of the given statements?  
(A) Only I follows  
(B) Only II follows

(C) I and III follows

(D) II and IV follows

**Directions (Qns. 35 to 37):** Each of the questions below is followed by two statements labelled (a) and (b) in which certain data are given. You are to determine whether the data given in the statement are sufficient for answering the question. Use the data given plus your knowledge of Mathematics and everyday facts to choose between four possible answers.

35. Is Chandrika the mother of Dabbu? To find out the answer to this question which of the following statements is/are sufficient?  
(a) Bhanu has two children of which Dabbu is one  
(b) Sister of Dabbu is daughter of Chandrika  
Codes:  
(A) Both (a) and (b) together are needed  
(B) Both (a) and (b) together are not sufficient  
(C) Only (a) is sufficient  
(D) Only (b) is sufficient
36. "No box is an animal". To reach this conclusion which of the following statements is/are sufficient?  
a) All boxes are ducks  
b) No animal is a duck  
Codes:  
(A) Both (a) and (b) together are needed  
(B) Both (a) and (b) together are not sufficient  
(C) Only (a) is sufficient  
(D) Only (b) is sufficient
37. "The tribals danced on Tuesday". To come to this conclusion, which of the following statements is/are sufficient?  
a) It rained on Tuesday  
b) When it rains the tribals usually dance  
Codes:  
(A) Both (a) and (b) together are needed  
(B) Both (a) and (b) together are not sufficient

- (C) Only (a) is sufficient  
(D) Only (b) is sufficient

Directions (Qns. 38 & 39): In the following questions, some statements are given followed by some conclusions. You have to decide which of the given conclusions follow from the given statements. You have to take the given statements to be true, even if they appear to be at variance with commonly known facts.

8. Statements:

- a) No Indian is Asian  
b) Some Americans are Indians

Conclusions :

- I Some Indians are not Asians  
II Some Asians are not Americans  
III Some Americans are not Asians  
IV All Americans are Asians

Codes:

- (A) Either III or IV follows  
(B) Either I or II follows  
(C) I and III follows  
(D) Either II or III follows

9. Statements:

- a) Some books are pencils  
b) No glasses are classes  
c) Some glasses are books

Conclusions :

- I Some glasses are pencils  
II Some books are not classes  
III Some books are classes

Codes:

- (A) Only I follows  
(B) Only II follows  
(C) Either II or III follows  
(D) Either I or II follows

Directions (Qns. 40 to 46): Choose the correct answer to the following questions.

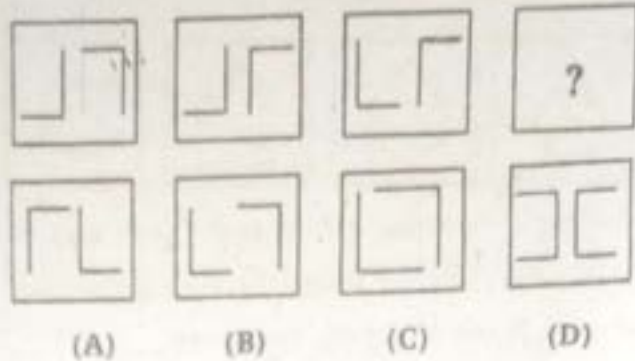
40. Simplify :

$$\frac{1.75 \times 1.75 \times 1.75 + 1.25 \times 1.25 \times 1.25}{1.75 \times 1.75 - 1.75 \times 1.25 + 1.25 \times 1.25}$$

- (A) 3  
(B) 1.25  
(C) 0.50  
(D) 1.75

41. Arrange the fractions  $\frac{3}{4}$ ,  $\frac{5}{7}$ ,  $\frac{9}{11}$ ,  $\frac{10}{13}$  in descending order  
(A)  $\frac{10}{13}$ ,  $\frac{9}{11}$ ,  $\frac{5}{7}$ ,  $\frac{3}{4}$   
(B)  $\frac{5}{7}$ ,  $\frac{3}{4}$ ,  $\frac{10}{13}$ ,  $\frac{9}{11}$   
(C)  $\frac{10}{13}$ ,  $\frac{5}{7}$ ,  $\frac{3}{4}$ ,  $\frac{9}{11}$   
(D)  $\frac{9}{11}$ ,  $\frac{10}{13}$ ,  $\frac{3}{4}$ ,  $\frac{5}{7}$

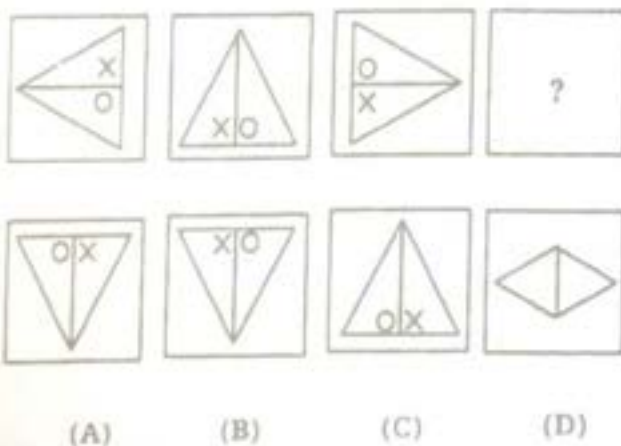
42.



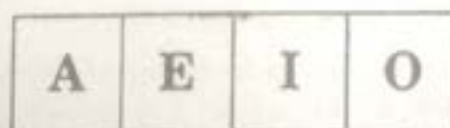
43. Which of the four boxes (A), (B), (C), (D) is most like the box on the left?



44.

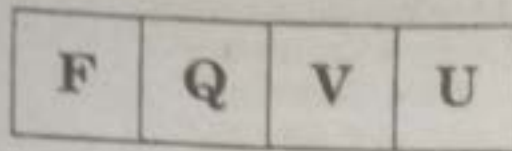


45. Select a related answer  
Problem Figures:



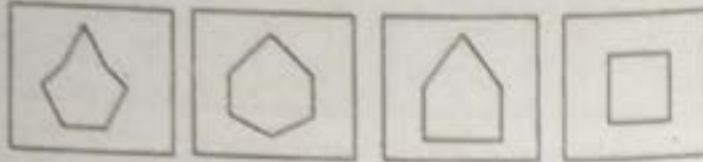


Answer Figures:



(A) (B) (C) (D)

46. Find the odd man out



(A) (B) (C) (D)

Directions (Qns. 47 to 49): Choose the correct answer to the following questions.

- A blue box weighs one-half of a red box
- A green box weighs 3 kgs more than a blue box
- A yellow box weighs the average of the weights of a green and red box.

47. Which colour box is the heaviest?

- (A) Blue  
(B) Green or red  
(C) Green  
(D) Red or yellow

48. Which colour box is the lightest?

- (A) Green  
(B) Red or blue  
(C) Blue  
(D) Green or red

49. If the blue box is 15 kg how much does the yellow box weigh?

- (A) 12 kg  
(B) 18 kg  
(C) 21 kg  
(D) 14 kg

Directions (Qns. 50 to 54): Choose the correct answer to the following questions.

50. If the sales tax on an appliance priced at Rs.300 is between 5 percent and 8 percent, then the cost (price plus sales tax) of the appliance could be

- (A) Rs.315  
(B) Rs.308  
(C) Rs.318  
(D) Rs.340

51. In a certain code language

- a) "so pa zu" "who is beautiful"  
b) "no zu to" means "she is thinking"

Then what is the code of "beautiful"?

- (A) so  
(B) pa  
(C) zu  
(D) so or pa

52. In this question, incomplete argument is followed by four statements, pick up one which will complete the argument.

People enjoy freedom in India because India is a democratic country.

- (A) In some democratic countries people enjoy freedom  
(B) In all democratic countries people enjoy freedom  
(C) In no democratic country people enjoy freedom  
(D) No one enjoy freedom in a democratic country

53. Which of the following does not belong to the group?

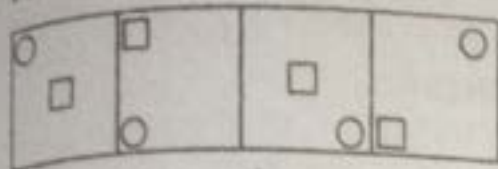
- (A) MP  
(B) RU  
(C) AD  
(D) HL

54. Which of the following answer choices would complete the series given below?

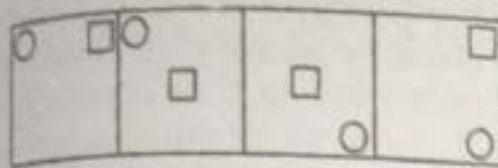
LNO PQS TVW XYA

- (A) BCD  
(B) BCE  
(C) BDE  
(D) DBF

Directions (Qns. 55 to 57): In the next three questions, select one of the 4 answers which should come after the problem figure if the sequence continues.

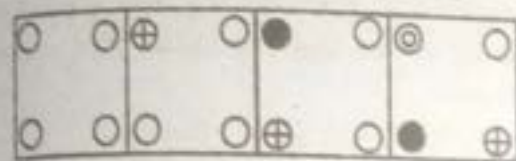


Answer Figures

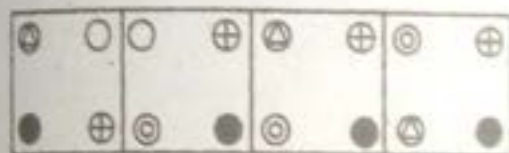


(A) (B) (C) (D)

56. Problem Figures



Answer Figures



(A) (B) (C) (D)

57. Problem Figures



Answer Figures



(A) (B) (C) (D)

Directions (Qns. 58 to 90): Choose the correct answer to the given questions.

58. In this question three statements are given, followed by three conclusions. Study the three statements and decide which of the conclusions follow:

- a) All cows are bulls.  
b) All elephants are ducks

c) No bulls are ducks

Conclusions :

- I No cows are ducks  
II No elephants are bulls  
III No elephants are cows

Codes:

- (A) Only I follows  
(B) I and II follows  
(C) I and III follows  
(D) All follows

59. Given: He is a responsible person because he is a married man.

Which statement expresses a conclusion that logically follows from the given statement?

- (A) All married persons are responsible  
(B) No married person is responsible  
(C) Many married persons are responsible  
(D) Some married persons are responsible

60. In a certain code a **TORMENTED** is written **NEMTORDET**, then how will **COMMUNIST** be written in the code?

- (A) NUMCOMIST  
(B) NUMMOCIST  
(C) NUMMOCTSI  
(D) NUMCOMTSI

61. The octal number corresponding to the binary number 11101.11 is

- (A) 35.6  
(B) 64.6  
(C) 64.3  
(D) 61.6

62. The simplified form of the Boolean expression  $y'z' + x'y' = z$ . Where Z is

- (A)  $z'$   
(B)  $z' + x'$   
(C)  $y'z' + x'$   
(D)  $y'$

63. Find the odd one out

- (A) Mouse  
(B) Keyboard  
(C) Touch-screen  
(D) Printer

64. One GB is equal to

- (A) 1000 MB  
(B) 100 MB



- (C) 1024 MB  
(D) 10 MB
65. ISDN stands for  
(A) International Standard Digital Network  
(B) International Subscriber Dialing Network  
(C) Integrated Service Digital Network  
(D) Integrated System Digital Network
66. India's first super computer is  
(A) Param  
(B) Flow solver  
(C) Trisul  
(D) Agni
67. NASSCOM stands for  
(A) National Association of Software Services Companies  
(B) National Association of System and Software Companies  
(C) National Software Services for Commercial Organisation  
(D) National Academy for System Simulation and Commercial Application
68. Which one of the following binary numbers represents hexadecimal number AB?  
(A) 10101011  
(B) 10111010  
(C) 11000011  
(D) 10101010
69. Consider the following program segment  
 $I = 0$   
 $L = 2$   
 Do 10 I = 1, 3  
 $10 J = (1+I) * L$   
 PRINT \*, I  
 The output is  
 (A) 12  
 (B) 14  
 (C) 3  
 (D) 8
70. The program segment  
 $PROD = 1$   
 DO 20 L = 1, 10  
 $DO 20 M = L, 10$   
 $PROD = PROD * A(L, M)$  20  
 CONTINUE  
 (A) Multiplies all the elements of the  $10 \times 10$  matrix A  
 (B) Multiplies all the elements below the leading diagonal of the  $10 \times 10$  matrix A  
 (C) Multiplies all the elements above the leading diagonal of the  $10 \times 10$  matrix A  
 (D) Multiplies all the elements other than the elements below the leading diagonal of the  $10 \times 10$  matrix A
71. What are the truth values of the boolean function F realized by the above logic circuit when the variables take values as follows:  
 a)  $A=0, B=0, C=1, D=1$   
 b)  $A=1, B=1, C=0, D=0$   
 Codes:  
 (A) (a) 0 (b) 1  
 (B) (a) 0 (b) 0  
 (C) (a) 1 (b) 1  
 (D) (a) 1 (b) 0
72. Java was originally invented by  
 (A) Microsoft  
 (B) Novell  
 (C) Sun  
 (D) Oracle
73. The BASIC Program statement corresponding to  $A+2B=D$  is  $\sqrt{C} + E^2$   
 (A)  $D=A+2*B$  (SQRT(C)+E\*E)  
 (B)  $A+2*B$  SQRT (C) + E\*E = D  
 (C)  $A+2*B$  SQRT (C)+E+E = D  
 (D)  $D=A+2*B$  SQRT (C)+E\*E
74. Which of the following is NOT operating system?  
 (A) Dos  
 (B) Unix  
 (C) Window NT  
 (D) Java
75. VSNL stands for  
 (A) Videsh Sanchar Nigam Ltd.  
 (B) Very Small Network Link  
 (C) Very Soft Network Link  
 (D) Vigam System Nigar Ltd.



76. Find the odd one out  
 (A) ORACLE  
 (B) SYBASE  
 (C) INFORMIX  
 (D) C
77. Pick the odd one out  
 (A) Microsoft  
 (B) IBM  
 (C) Novell  
 (D) Windows
78. CD-ROM is a/an  
 (A) Optical device  
 (B) Magnetic device  
 (C) Electro Magnetic Device  
 (D) Mechanical Device
79. The capacity of 3½" Floppy Disk is  
 (A) 1.2 MB  
 (B) 1.4 MB  
 (C) 1.44 MB  
 (D) 1 MB
80. ORDBMS stands for  
 (A) Open Relational Database Management System  
 (B) Object Relational Database Management System  
 (C) Other Relational Database Management System  
 (D) Object Relational Database Maintenance System
81. Which of the following memory helps computer for fast computation?  
 (A) Cache  
 (B) ROM  
 (C) RAM  
 (D) CD-ROM
82. Liquid hydrocarbons are converted to low molecular weight gaseous hydrocarbons by a process known as  
 (A) Hydrogenation  
 (B) Reforming  
 (C) Cracking  
 (D) Reduction
83. The largest stable nucleus is  
 (A) U-238  
 (B) U-235  
 (C) Pb-206  
 (D) Bi-209
84. Human Immune virus (HIV) is known to cause  
 (A) Blood cancer  
 (B) AIDS  
 (C) Cerebral haemorrhage  
 (D) Hepatitis
85. What is the Kinetic Energy of a static particle with mass "m"?  
 (A) mgh  
 (B)  $mgh + \frac{1}{2} mv^2$   
 (C) 0  
 (D)  $\infty$
86. The principle of LASER is  
 (A) Stimulated Absorption  
 (B) Stimulated Emission  
 (C) Spontaneous Absorption  
 (D) Spontaneous Emission
87. The wave length range of X-rays is  
 (A) 3000–7000 Å  
 (B) 200–500 Å  
 (C) 0.1–10 Å  
 (D) 0.01–0.1 Å
88. Which is the semiconductor element?  
 (A) Copper  
 (B) Aluminium  
 (C) Silicon  
 (D) Bismuth
89. pH value of distilled water is  
 (A) 9  
 (B) 7  
 (C) 8  
 (D) 13
90. Frequency range of the audio signal is  
 (A) above 20 KHz  
 (B) below 20 Hz  
 (C) 20 Hz–20 KHz  
 (D) 3 MHz–30 MHz
- Directions (Qns. 91 & 92): In the next two questions select the word or phrase whose meaning is closest to the given in capital letter*
91. CONDONE  
 (A) express sympathy  
 (B) forget

(C) revenge

(D) forgive

92. **PONDER**

(A) reflect

(B) hazard

(C) reject

(D) think

*Directions (Qns. 93 & 94): Which of the words following the capitalised word is opposite in meaning to the capitalised word?*

93. **FANATICISM**

(A) perplexity

(B) excessive zeal

(C) excessive enthusiasm

(D) endurance

94. **OUST**

(A) evict

(B) banish

(C) expel

(D) admit

*Directions (95 to 97): Select the pair of words which are related in the same way as the capitalised words are related to each other.*

95. **HOSPITALITY : HOST**

(A) Expatriate : Exile

(B) Asylum : Prison

(C) Prisoner : Prison

(D) Board : Parole

96. **COPY : ORIGINAL**

(A) Several : One

(B) Image : Photograph

(C) Photostat : Document

(D) Telegraph : Transmission

97. **ICE : WATER**

(A) Rain : Snow

(B) Plane : Safety Belt

(C) Sky : Clouds

(D) Wood : Sap

*Directions (Qns. 98 to 100): Each question consists of a sentence with three words or phrases underlined. Select that word/phrase which contains an error in usage, grammar or punctuation. If there is no error write 4 as the answer.*

98. The news is very good and it is hard to  
1 2

be believed.

3

99. None of the two boys could tell the truth.  
1 2 3

100. Many workers feel that there are  
1 2

too much rules in the company.

3



**ANSWERS**

|        |        |        |        |        |        |        |        |        |         |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| 1 (B)  | 2 (A)  | 3 (B)  | 4 (C)  | 5 (B)  | 6 (C)  | 7 (C)  | 8 (A)  | 9 (C)  | 10 (D)  |
| 11 (B) | 12 (C) | 13 (A) | 14 (B) | 15 (C) | 16 (B) | 17 (C) | 18 (B) | 19 (B) | 20 (B)  |
| 21 (B) | 22 (B) | 23 (C) | 24 (B) | 25 (D) | 26 (C) | 27 (B) | 28 (C) | 29 (B) | 30 (B)  |
| 31 (B) | 32 (C) | 33 (B) | 34 (C) | 35 (D) | 36 (A) | 37 (A) | 38 (D) | 39 (A) | 40 (A)  |
| 41 (D) | 42 (B) | 43 (C) | 44 (A) | 45 (D) | 46 (D) | 47 (C) | 48 (B) | 49 (D) | 50 (A)  |
| 51 (B) | 52 (B) | 53 (D) | 54 (C) | 55 (B) | 56 (C) | 57 (A) | 58 (D) | 59 (A) | 60 (D)  |
| 61 (A) | 62 (B) | 63 (D) | 64 (A) | 65 (C) | 66 (A) | 67 (A) | 68 (A) | 69 (B) | 70 (C)  |
| 71 (B) | 72 (C) | 73 (A) | 74 (D) | 75 (A) | 76 (D) | 77 (D) | 78 (A) | 79 (C) | 80 (B)  |
| 81 (C) | 82 (D) | 83 (A) | 84 (B) | 85 (B) | 86 (B) | 87 (C) | 88 (C) | 89 (B) | 90 (C)  |
| 91 (D) | 92 (D) | 93 (D) | 94 (D) | 95 (C) | 96 (B) | 97 (C) | 98 (C) | 99 (D) | 100 (B) |

